HIMANSHU SHUKLA

SOFTWARE ENGINEERING | BACKEND DEVELOPMENT

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PROFESSIONAL SUMMARY

Mathematics & Computing undergraduate developing API-backed applications, data pipelines and testable software. Projects include assessment workflows, document crawling, network monitoring, real-time collaboration and governance dashboards.

EDUCATION

Madhav Institute of Technology & Science (MITS), Gwalior | B.Tech, Mathematics & Computing
Expected graduation: 2028 | CGPA: 7.75/10 (through Semester IV)

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, TypeScript, SQL
Backend and databases: FastAPI, REST APIs, PostgreSQL, Supabase, Firebase, Firestore, Alembic
Testing and tools: pytest, Docker, Git/GitHub, Linux command line, Vercel, Jupyter Notebook
Other technologies: React, Next.js, Tailwind CSS, Scapy, PyTorch, FAISS

PROJECT EXPERIENCE

Mulyankan | Assessment and reviewer workflow platform | 3D explanation
React, TypeScript, Supabase | Demonstrated prototype

- Developed submission, evaluator dashboards, rubric scoring and tracking workflows, including evidence-linked review and human override/audit controls.
- Presented the project at CMPDI headquarters in Ranchi (July 2026) following institute-level Smart India Hackathon selection.

NewsSources | News and PDF ingestion system
Python, PostgreSQL, Docker, Alembic | In development

- Implemented RSS/sitemap discovery and crawling components for article collection, source tracking and URL deduplication.
- Designed a PostgreSQL-backed ingestion workflow for full articles and PDFs, with versioned database migrations and containerized services.

OHS (Our Hidden Universe) | Real-time collaboration web application | ohs.vercel.app
React, Firebase Authentication, Firestore, Vercel | Deployed

- Built private-room authentication, real-time messaging, video calling and shared media with Firebase-backed persistence.
- Implemented synchronized Watch Together playback and real-time shared-content updates.

Sentinel | Network monitoring and filtering system
Python, FastAPI, Scapy | Locally tested prototype

- Developed packet/flow monitoring API views and an initial anomaly-analysis interface; validated forward-proxy domain blocking with local HTTP tests.

Coal Governance | Inspection and compliance dashboards
Next.js, PostgreSQL, MapLibre, Flyway | Prototype

- Developed GIS-linked views for inspections, violations and corrective actions; validated versioned database migrations through V026.

ADDITIONAL PROJECTS

EarthBridge (PyTorch/FAISS retrieval, FastAPI demo); IrisTrack (webcam gaze/blink and calibration); Mulyankan Spatial Lab (interactive 3D explainer).

LEADERSHIP & ACHIEVEMENTS

Class Representative (3rd Year); NPTEL Coordinator; Training & Placement Cell Member; SchoolBuddies co-founder. First Position, Neeti Summit English Debate; DevFusion 4.0 (advanced to Round 3).

RELEVANT COURSEWORK

Data Structures and Algorithms, Database Management Systems, Operating Systems, Design and Analysis of Algorithms, Data Science.